

STA-121 STATISTICAL INFERENCE

Q1. Which of the following is a point estimate?

- A. Confidence interval
- B. Sample mean
- C. Margin of error
- D. Population parameter

Answer: B. Sample mean

Q2. In hypothesis testing, the null hypothesis (H_0) usually represents:

- A. No effect or no difference
- B. The research claim
- C. Always true assumption
- D. Alternative claim

Answer: A. No effect or no difference

Q3. A population is best described as:

- A. The entire set of individuals or items of interest
- B. A part of the whole group
- C. A sampling frame
- D. An estimate of the mean

Answer: A. The entire set of individuals or items of interest

Q4. In regression analysis, the dependent variable is also called:

- A. Predictor variable
- B. Explanatory variable
- C. Response variable
- D. Independent variable

Answer: C. Response variable

Q5. The margin of error in a confidence interval depends on:

- A. Sample size and standard error
- B. Population size only
- C. Median of data
- D. Mode of data

Answer: A. Sample size and standard error

Q6. The probability of rejecting a true null hypothesis is:

- A. Power of the test
- B. Type I error
- C. Type II error
- D. Confidence level

Answer: B. Type I error

Q7. A sample is defined as:

- A. Entire data set
- B. Portion of the population selected for analysis
- C. The same as population
- D. Population parameter

Answer: B. Portion of the population selected for analysis

Q8. If correlation coefficient $r = 0.95$, the relationship is:

- A. Very weak positive
- B. Strong positive
- C. Weak negative
- D. Strong negative

Answer: B. Strong positive

Q9. A 95% confidence interval implies:

- A. The parameter will fall in the interval 95% of the time
- B. 95% of the data lies in the interval
- C. 95% of samples will include the population parameter
- D. Both A and C

Answer: C. 95% of samples will include the population parameter

Q10. A two-tailed test is used when:

- A. The researcher is only interested in increases
- B. The researcher is only interested in decreases
- C. The researcher is interested in differences in both directions
- D. H_0 is always accepted

Answer: C. The researcher is interested in differences in both directions

Q11. Which of the following is a parameter?

- A. Sample mean
- B. Population mean
- C. Sample variance
- D. Sample proportion

Answer: B. Population mean

Q12. In simple linear regression, the slope coefficient represents:

- A. Change in y per unit change in x
- B. Intercept on the y-axis
- C. Error term
- D. Correlation strength

Answer: A. Change in y per unit change in x

Q13. A point estimator should ideally be:

- A. Biased and consistent
- B. Unbiased and efficient
- C. Large and skewed

D. Variable and random

Answer: B. Unbiased and efficient

Q14. The p-value in hypothesis testing represents:

A. The probability H_0 is true

B. The probability of observing the sample data if H_0 is true

C. The significance level α

D. The power of the test

Answer: B. The probability of observing the sample data if H_0 is true

Q15. Stratified random sampling involves:

A. Dividing population into groups and selecting proportionally

B. Selecting every nth element

C. Using volunteers

D. Randomly picking clusters

Answer: A. Dividing population into groups and selecting proportionally

Q16. A correlation coefficient close to 0 indicates:

A. Strong linear relationship

B. No linear relationship

C. Perfect correlation

D. Negative relationship

Answer: B. No linear relationship

Q17. The width of a confidence interval decreases when:

A. Sample size increases

B. Sample variance increases

C. Confidence level increases

D. All of the above

Answer: A. Sample size increases

Q18. A one-tailed test is appropriate when:

A. The research hypothesis specifies a direction

B. The hypothesis is non-directional

C. The null hypothesis is false

D. The sample is large

Answer: A. The research hypothesis specifies a direction

Q19. The statistic used to estimate population proportion is:

A. p (sample proportion)

B. μ (population mean)

C. σ (population standard deviation)

D. $\bar{x}$ (sample mean)

Answer: A. p (sample proportion)

Q20. In regression, R^2 measures:

- A. Slope of the regression line
- B. Correlation between x and y
- C. Proportion of variance in y explained by x
- D. Standard error of estimate

Answer: C. Proportion of variance in y explained by x

Q21. Confidence intervals are typically expressed as:

- A. Point estimate $\pm$ margin of error
- B. Variance of population
- C. Standard error $\times$ n
- D. Mode $\pm$ frequency

Answer: A. Point estimate $\pm$ margin of error

Q22. The critical region of a test is:

- A. Values of the test statistic that lead to rejection of H_0
- B. Always the right tail
- C. Where p-value $> \alpha$
- D. Region where H_0 is true

Answer: A. Values of the test statistic that lead to rejection of H_0

Q23. Sampling error occurs because:

- A. We observe only a subset of the population
- B. We include everyone in the population
- C. The parameter equals the statistic
- D. Measurement errors only

Answer: A. We observe only a subset of the population

Q24. In regression, the error term captures:

- A. Variation in y not explained by x
- B. The slope of the line
- C. The intercept of the line
- D. Correlation coefficient

Answer: A. Variation in y not explained by x

Q25. If sample size increases, the standard error:

- A. Decreases
- B. Increases
- C. Remains constant
- D. Doubles

Answer: A. Decreases

Q26. The alternative hypothesis (H_1) usually states:

- A. There is an effect or difference
- B. No effect exists
- C. That H_0 is always true

D. Sample mean = population mean

Answer: A. There is an effect or difference

Q27. The mean of a sampling distribution of the sample mean equals:

A. Population mean

B. Sample variance

C. Population variance

D. Sample standard deviation

Answer: A. Population mean

Q28. The correlation coefficient (r) lies between:

A. 0 and 1

B. -1 and +1

C. $-\infty$ and ∞

D. 0 and ∞

Answer: B. -1 and +1

Q29. In estimation, the confidence coefficient is denoted by:

A. $(1 - \alpha)$

B. β

C. σ^2

D. μ

Answer: A. $(1 - \alpha)$

Q30. If $p\text{-value} < \alpha$, we:

A. Reject H_0

B. Fail to reject H_0

C. Accept H_0

D. Increase sample size

Answer: A. Reject H_0

Q31. The main purpose of estimation is to:

A. Infer population parameters from sample statistics

B. Increase sample size

C. Eliminate bias entirely

D. Ensure sampling error is zero

Answer: A. Infer population parameters from sample statistics

Q32. Type II error occurs when we:

A. Reject H_0 when H_0 is true

B. Fail to reject H_0 when H_0 is false

C. Reject H_1 when it is true

D. Use wrong test statistic

Answer: B. Fail to reject H_0 when H_0 is false

Q33. Which of these is NOT a sampling method?

- A. Simple random sampling
- B. Stratified sampling
- C. Cluster sampling
- D. Arithmetic sampling

Answer: D. Arithmetic sampling

Q34. In regression, the intercept represents:

- A. Value of y when $x = 0$
- B. Value of x when $y = 0$
- C. Slope of regression line
- D. Standard error of estimate

Answer: A. Value of y when $x = 0$

Q35. The wider a confidence interval, the:

- A. Greater the confidence level
- B. Lower the confidence level
- C. More precise the estimate
- D. Less uncertainty

Answer: A. Greater the confidence level

Q36. At significance level $\alpha = 0.05$, what is the probability of a Type I error?

- A. 0.95
- B. 0.05
- C. 0.50
- D. Cannot be determined

Answer: B. 0.05

Q37. A statistic is called unbiased if:

- A. Its expected value equals the parameter it estimates
- B. Its variance is zero
- C. It equals the population value in every sample
- D. It is always normally distributed

Answer: A. Its expected value equals the parameter it estimates

Q38. Correlation measures:

- A. Cause-and-effect
- B. Strength and direction of linear relationship
- C. Sampling variability
- D. Error variance

Answer: B. Strength and direction of linear relationship

Q39. A 99% confidence interval is:

- A. Wider than a 95% confidence interval
- B. Narrower than a 95% confidence interval
- C. Equal in width to a 95% confidence interval

D. Independent of sample size

Answer: A. Wider than a 95% confidence interval

Q40. The power of a statistical test is:

A. $1 - \beta$

B. α

C. $1 - \alpha$

D. β

Answer: A. $1 - \beta$

Q41. Systematic sampling selects units by:

A. Randomly choosing every k th element

B. Dividing population into strata

C. Randomly selecting clusters

D. Volunteer participation

Answer: A. Randomly choosing every k th element

Q42. In regression, multicollinearity refers to:

A. Independent variables being highly correlated

B. Dependent variable being categorical

C. Error term being normally distributed

D. Having more than one dependent variable

Answer: A. Independent variables being highly correlated

Q43. An estimator with smaller variance compared to others is called:

A. Efficient

B. Biased

C. Consistent

D. Random

Answer: A. Efficient

Q44. A left-tailed test is used when the research hypothesis states:

A. Population parameter is less than a specific value

B. Population parameter is greater than a specific value

C. Population parameter is not equal to a specific value

D. None of the above

Answer: A. Population parameter is less than a specific value

Q45. Which of these is an example of a population parameter?

A. Population variance

B. Sample variance

C. Sample proportion

D. Sample mean

Answer: A. Population variance

Q46. In regression, residuals are defined as:

- A. Observed value minus predicted value
- B. Predicted value minus observed value
- C. Total variation in y
- D. Error variance only

Answer: A. Observed value minus predicted value

Q47. Larger samples generally lead to:

- A. Smaller sampling error
- B. Larger confidence intervals
- C. More biased estimators
- D. Greater variability in estimates

Answer: A. Smaller sampling error

Q48. If $\alpha = 0.01$, this means:

- A. 1% chance of committing Type I error
- B. 10% chance of committing Type I error
- C. 99% chance of rejecting H_0
- D. 1% chance of committing Type II error

Answer: A. 1% chance of committing Type I error

Q49. In probability sampling:

- A. Every unit has a known chance of selection
- B. Only volunteers are used
- C. Selection is based on judgment
- D. Clusters are ignored

Answer: A. Every unit has a known chance of selection

Q50. In regression, the method of least squares minimizes:

- A. Sum of squared residuals
- B. Sum of residuals
- C. Variance of x
- D. Standard deviation of y

Answer: A. Sum of squared residuals

Q51. Point estimation uses:

- A. A single value to estimate a parameter
- B. A range of values to estimate a parameter
- C. Confidence levels
- D. Standard error only

Answer: A. A single value to estimate a parameter

Q52. A test is called robust if:

- A. It gives valid results even when assumptions are violated
- B. It never makes errors
- C. It always rejects H_0

D. It is non-parametric

Answer: A. It gives valid results even when assumptions are violated

Q53. The central limit theorem states that the sampling distribution of the mean:

A. Approaches normality as sample size increases

B. Is always skewed

C. Equals the sample distribution

D. Equals the population distribution

Answer: A. Approaches normality as sample size increases

Q54. In correlation, $r = -0.8$ indicates:

A. Strong negative linear relationship

B. Weak negative relationship

C. Strong positive relationship

D. No relationship

Answer: A. Strong negative linear relationship

Q55. Interval estimation is preferred over point estimation because:

A. It provides a range likely to contain the parameter

B. It gives exact population values

C. It eliminates sampling error

D. It is always unbiased

Answer: A. It provides a range likely to contain the parameter

Q56. The null hypothesis is rejected if:

A. Test statistic falls in the critical region

B. $P\text{-value} > \alpha$

C. Confidence interval includes hypothesized parameter

D. Both B and C

Answer: A. Test statistic falls in the critical region

Q57. Which sampling method is used when population is geographically scattered?

A. Cluster sampling

B. Stratified sampling

C. Systematic sampling

D. Convenience sampling

Answer: A. Cluster sampling

Q58. In regression, adjusted R^2 is preferred when:

A. Comparing models with different numbers of predictors

B. Only one predictor exists

C. Sample size is infinite

D. Errors are zero

Answer: A. Comparing models with different numbers of predictors

Q59. An estimator is consistent if:

- A. It converges to the true parameter as sample size increases
- B. Its variance is zero
- C. It is always unbiased
- D. It equals the sample statistic

Answer: A. It converges to the true parameter as sample size increases

Q60. A researcher sets $\alpha = 0.05$. This means:

- A. 5% chance of rejecting H_0 when it is true
- B. 95% chance of rejecting H_0 when it is false
- C. 5% chance of committing Type II error
- D. 95% confidence interval

Answer: A. 5% chance of rejecting H_0 when it is true

Q61. A 95% confidence interval means:

- A. The parameter is always inside the interval
 - B. 95% of such intervals from repeated samples contain the parameter
 - C. The sample mean is 95% accurate
 - D. There is a 95% chance the parameter moves
- Answer: B. 95% of such intervals from repeated samples contain the parameter**

Q62. The null hypothesis (H_0) is usually stated as:

- A. No difference or no effect
- B. The effect always exists
- C. The parameter is unknown
- D. The research claim

Answer: A. No difference or no effect

Q63. In sampling, the finite population correction is applied when:

- A. Sample size is more than 5% of the population
- B. Sample size is less than 5% of the population
- C. Population is infinite
- D. Sample is biased

Answer: A. Sample size is more than 5% of the population

Q64. The correlation coefficient $r = 0$ means:

- A. No linear relationship
- B. No relationship at all
- C. Strong negative correlation
- D. Strong positive correlation

Answer: A. No linear relationship

Q65. If the test statistic falls in the acceptance region, we:

- A. Fail to reject H_0
- B. Reject H_0
- C. Increase α

D. Change the hypothesis

Answer: A. Fail to reject H_0

Q66. A point estimate is always:

- A. A single value
- B. A range of values
- C. Equal to the population parameter
- D. A biased statistic

Answer: A. A single value

Q67. Which of the following is NOT a property of a good estimator?

- A. Unbiasedness
- B. Consistency
- C. Efficiency
- D. Complexity

Answer: D. Complexity

Q68. In regression, R^2 represents:

- A. Proportion of variation in y explained by x
- B. Slope of the line
- C. Error variance
- D. Standard deviation of x

Answer: A. Proportion of variation in y explained by x

Q69. The smaller the p-value:

- A. Stronger the evidence against H_0
- B. Weaker the evidence against H_0
- C. Stronger the evidence for H_0
- D. Larger the confidence interval

Answer: A. Stronger the evidence against H_0

Q70. The sample proportion is an estimator of:

- A. Population proportion
- B. Population variance
- C. Population mean
- D. Sample variance

Answer: A. Population proportion

Q71. Type I error occurs when we:

- A. Reject H_0 when H_0 is true
- B. Fail to reject H_0 when H_0 is false
- C. Reject H_1 when H_1 is true
- D. Use wrong test statistic

Answer: A. Reject H_0 when H_0 is true

Q72. The slope in a regression equation shows:

- A. Change in y for a unit change in x
- B. Change in x for a unit change in y
- C. Value of y when $x = 0$
- D. Standard error of estimate

Answer: A. Change in y for a unit change in x

Q73. A 99% confidence level means the probability of Type I error is:

- A. 0.01
- B. 0.05
- C. 0.10
- D. 0.99

Answer: A. 0.01

Q74. A parameter refers to:

- A. Numerical characteristic of a population
- B. Numerical characteristic of a sample
- C. Statistic computed from data
- D. Random variable

Answer: A. Numerical characteristic of a population

Q75. If $r = 0.95$, this implies:

- A. Strong positive linear relationship
- B. Weak positive relationship
- C. No relationship
- D. Negative relationship

Answer: A. Strong positive linear relationship

Q76. A two-tailed test is appropriate when:

- A. We are interested in deviations on both sides of the mean
- B. Only values greater than mean matter
- C. Only values less than mean matter
- D. Variance is unknown

Answer: A. We are interested in deviations on both sides of the mean

Q77. The standard error of the mean is calculated as:

- A. $\sigma / \sqrt{n}$
- B. σ^2 / n
- C. $\mu / \sqrt{n}$
- D. n / σ

Answer: A. $\sigma / \sqrt{n}$

Q78. In regression, if the slope = 0, it means:

- A. No linear relationship
- B. Strong negative relationship
- C. Strong positive relationship

D. Errors are zero

Answer: A. No linear relationship

Q79. A consistent estimator becomes:

A. Closer to the parameter as sample size increases

B. Always equal to the parameter

C. Less reliable with larger samples

D. Independent of sample size

Answer: A. Closer to the parameter as sample size increases

Q80. The probability of failing to reject H_0 when it is false is:

A. β

B. α

C. $1 - \alpha$

D. $1 - \beta$

Answer: A. β

Q81. In cluster sampling, the population is divided into:

A. Groups, and some groups are randomly selected

B. Groups, and elements from all groups are selected

C. Equal parts only

D. Strata of equal size

Answer: A. Groups, and some groups are randomly selected

Q82. In regression, residual sum of squares measures:

A. Variation in y not explained by regression

B. Variation in y explained by regression

C. Variation in x

D. Total variation in y

Answer: A. Variation in y not explained by regression

Q83. The larger the sample size, the confidence interval becomes:

A. Narrower

B. Wider

C. Constant

D. More biased

Answer: A. Narrower

Q84. A right-tailed test is used when:

A. Research hypothesis states population parameter $>$ hypothesized value

B. Research hypothesis states population parameter $<$ hypothesized value

C. Research hypothesis states population parameter $\neq$ hypothesized value

D. Population is normal

Answer: A. Research hypothesis states population parameter $>$ hypothesized value

Q85. The mean of sample means equals:

- A. Population mean
- B. Sample variance
- C. Sample proportion
- D. Population variance

Answer: A. Population mean

Q86. Correlation does not imply:

- A. Causation
- B. Linear relationship
- C. Strength of relationship
- D. Direction of relationship

Answer: A. Causation

Q87. The margin of error in interval estimation depends on:

- A. Critical value and standard error
- B. Only population variance
- C. Only sample size
- D. Only population size

Answer: A. Critical value and standard error

Q88. If p-value = 0.20 and $\alpha = 0.05$, we:

- A. Fail to reject H_0
- B. Reject H_0
- C. Accept H_1
- D. Increase α

Answer: A. Fail to reject H_0

Q89. Stratified sampling is useful when:

- A. Population has distinct subgroups
- B. Population is homogeneous
- C. Randomization is impossible
- D. Only one variable exists

Answer: A. Population has distinct subgroups

Q90. In regression, the coefficient of determination (R^2) = 0 means:

- A. No variation in y is explained by x
- B. All variation in y is explained by x
- C. Errors are zero
- D. Slope = 1

Answer: A. No variation in y is explained by x

Q91. The standard normal distribution has mean =

- A. 0 and variance = 1
- B. 1 and variance = 0
- C. 0 and variance = 0

D. 1 and variance = 1

Answer: A. 0 and variance = 1

Q92. If sample size is small and σ unknown, we use:

- A. t-distribution
- B. z-distribution
- C. Poisson distribution
- D. Chi-square distribution

Answer: A. t-distribution

Q93. A biased estimator means:

- A. Expected value $\neq$ true parameter
- B. Expected value = true parameter
- C. Variance = 0
- D. It is efficient

Answer: A. Expected value $\neq$ true parameter

Q94. In correlation, r close to 0 means:

- A. Very weak linear relationship
- B. Strong linear relationship
- C. No variability in data
- D. High causation

Answer: A. Very weak linear relationship

Q95. The total sum of squares in regression equals:

- A. Regression SS + Residual SS
- B. Residual SS only
- C. Regression SS only
- D. Error SS only

Answer: A. Regression SS + Residual SS

Q96. The significance level (α) is set by:

- A. Researcher
- B. Population
- C. Sample
- D. Computer

Answer: A. Researcher

Q97. The law of large numbers states:

- A. Sample mean approaches population mean as n increases
- B. Sample variance approaches 0 as n increases
- C. Population mean equals sample mean always
- D. Probability approaches infinity

Answer: A. Sample mean approaches population mean as n increases

Q98. In hypothesis testing, the test statistic is compared with:

- A. Critical value
- B. Sample size
- C. Standard error
- D. Population mean

Answer: A. Critical value

Q99. If correlation = -1 , then:

- A. Perfect negative linear relationship
- B. Perfect positive linear relationship
- C. No relationship
- D. Weak negative relationship

Answer: A. Perfect negative linear relationship

Q100. A sample statistic is used to:

- A. Estimate population parameter
- B. Replace parameter completely
- C. Avoid sampling error
- D. Eliminate random variation

Answer: A. Estimate population parameter